

Lecture 11

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Sliding mode control and its application

Outline

- 1 Design of Sliding Mode Control
- 2 Sliding Mode Design Using Full Order Lyapunov Matrix
- 3 Sliding Hyperplane Design Under Parametric Uncertainties

Sliding Mode Control

Consider a single-input system

$$\dot{x} = Ax + Bu + Bd, \quad x \in \mathbb{R}^n, \quad u \in \mathbb{R}$$

and the sliding manifold

$$S = \{x \in \mathbb{R}^n : s = c^T x = 0\}.$$

- Since in this case only one control input appears, here sliding variable $s = c^T x$ is a real-valued function.
- Once the sliding mode is enforced in the system, $s = 0$, and the trajectories are confined to S .
- Switching/discontinuous control become active on S .

Consider a multi-input system

$$\dot{x} = Ax + Bu + Bd, \quad x \in \mathbb{R}^n, \quad u \in \mathbb{R}^m$$

Note that there will be m number of sliding variables, i.e., $s \in \mathbb{R}^m$. So, the sliding manifold is given as

$$S = \{x \in \mathbb{R}^n : s = c^T x = 0\}$$

where $c \in \mathbb{R}^{n \times m}$.

What are the challenges in MIMO?

- Since there are m sliding variables, it may be possible that sliding mode may occur in i -th component of s , where $i = 1, 2, \dots, m$.
- Design a control which brings the sliding mode into the system.

Sliding Mode

Definition

The system is said to be in sliding mode if the system trajectories are confined to

$$\mathcal{S} = \bigcap_{i=1}^m \mathcal{S}_i$$

where the hyperplane $\mathcal{S}_i$ is given as

$$\mathcal{S}_i = \{x \in \mathbb{R}^n : s_i = c_i^\top x = 0\}.$$

Here, c_i is i -th column of c .

- Clearly, in multi-input case, the sliding mode is enforced only when all the components of s become zero or in other words, $s = 0$
- If $m = 1$, this definition reduces to the same as that defined in the single-input case
- In multi-input, it may happen sliding takes place in any i -th sliding hyperplane before sliding mode is enforced in the system

Sliding Mode Control

There are two schools of thought on the design of sliding mode control for MIMO systems.

Diagonalization Approach

- The matrix $c^T B$ is transformed into diagonal form first.
- Here, the control is designed such that each control affects only the respective sliding variable.
- Each control element brings the sliding mode on its respective sliding hyperplane, i.e., u_i forces the system trajectory to the manifold $\mathcal{S}_i$ and so on.
- Sliding mode occurs only when trajectories are forced to all m -th sliding hyperplanes.

Unit Vector Approach

- Switching control is designed such that its sliding mode is enforced to the manifold $\mathcal{S}$. It is given as

$$u_{\text{dis}} = \frac{s}{\|s\|}$$

for any $s \in \mathbb{R}^m$. For $m = 1$, it reduces to $\frac{s}{|s|}$ which is equivalent to $\text{sign}(s)$.

- Here, control switches only when the trajectories reach the manifold $\mathcal{S}$.
- Compared to the diagonalization approach, the switching signal does not affect when the trajectory crosses any of these sliding hyperplanes.

Diagonalization Approach

- Differentiating $s = c^\top x$ gives $\dot{s} = c^\top Ax + c^\top Bu + c^\top Bd$.
- The sliding surface parameter is chosen such that $c^\top B$ is a diagonal matrix.
- Sliding mode control can be derived as $u = -(c^\top B)^{-1} (c^\top Ax + K \text{signs})$.
- Here, K is a diagonal matrix with diagonal elements K_i for $i = 1, 2, \dots, m$ and $\text{signs} = [\text{sign}s_1 \ \cdots \ \text{sign}s_m]^\top$.
- Thus, it can be easily written component-wise as

$$u_i = -(c^\top B)_i^{-1} ((c^\top A)_i x + K_i \text{sign}s_i), \quad \forall i = 1, 2, \dots, m.$$

The closed-loop system becomes

$$\dot{s}_i = -K_i \text{sign}s_i + (c^\top B)_i d_i, \quad \forall i = 1, 2, \dots, m$$

where $K_i > \sup_{t \geq 0} |(c^\top B)_i d_i(t)|$. Hence, the sliding mode is enforced when all s_i 's are zero.

Unit Vector Approach

Sliding Mode Control

Consider again the sliding dynamics

$$\dot{s} = c^T Ax + c^T Bu + c^T Bd.$$

In order to ensure $s = 0$, design the control as

$$u = -(c^T B)^{-1} \left(c^T Ax + K \frac{s}{\|s\|} \right).$$

Here, the switching gain K is a scalar. The closed loop system stability can be obtained by substituting the control in $\dot{s}$ is given as

$$\dot{s} = -K \frac{s}{\|s\|} + c^T Bd.$$

The convergence of the system trajectory to the sliding manifold $\mathcal{S}$ is discussed now. This is possible only when $s = 0$ in finite time.

Unit Vector Approach

Theorem:

Sliding mode occurs in the system with defined control if $K > \|c^\top B\| d_m$.

Proof

Consider a Lyapunov function $V = s^\top s$. Taking the derivation of s along the system trajectory, we obtain

$$\begin{aligned}\dot{V} &= \dot{s}^\top s + s^\top \dot{s} \\ &= -K \frac{s^\top s}{\|s\|} + d^\top B^\top c s - K \frac{s^\top s}{\|s\|} + s^\top c^\top B d \\ &= -2K\|s\| + 2s^\top c^\top B d \\ &\leq -2K\|s\| + 2\|s\| \|c^\top B d\|.\end{aligned}$$

We have used the fact $s^\top s = \|s\|^2$ in third equality. If the d is bounded, i.e., $\|d\| < d_m$,

$$\begin{aligned}\dot{V} &\leq -2K\|s\| + 2\|s\| \|c^\top B\| d_m \\ &= -2(K - \|c^\top B\| d_m) V^{1/2} \\ &\leq -2\eta V^{1/2}\end{aligned}$$

where $\eta > 0$.

Unit Vector Approach

Proof

Integrating both sides of the above differential inequality with the condition $V(t_1) = 0$ gives

$$\int_{V_0}^0 V^{-1/2} dV \leq -2\eta \int_0^{t_1} dt \implies -2V_0^{1/2} \leq -2\eta t_1.$$

Thus, the system trajectory reaches the sliding manifold in a finite time not larger than $t_1 \leq \frac{V_0^{1/2}}{\eta}$.

Hence, the unit vector approach to sliding mode control ensures the sliding mode in the system and the proof is completed.

Sliding Mode Design Using Full Order Lyapunov Matrix

System Description

We consider a system in regular form

$$\dot{x}_1 = A_{11}x_1 + A_{12}x_2$$

$$\dot{x}_2 = A_{21}x_1 + A_{22}x_2 + B_2u + B_2d.$$

Here, we consider a multi-input system, i.e., $u \in \mathbb{R}^m$. Though this design technique is explicitly stated for multi-input, it is easily extended to the single-input case with minor modifications.

Control Problems

In the following, we answer to the following questions:

- Can we design sliding variable (s) using the full order system rather than the reduced dynamics?
- Is it possible to characterize the system behaviour using full order dynamics of the system?

Sliding Mode Design Using Full Order Lyapunov Matrix

Review

Consider sliding function $s = c^T x = c_1^T x_1 + x_2$.

- We know that there exists a control (sliding mode) law such that $s = 0$ in finite time. It implies $x_2 = -c_1^T x_1$.
- Substituting this in the upper part of system dynamics,

$$\dot{x}_1 = (A_{11} - A_{12}c_1^T) x_1.$$

- Thus, the closed-loop system dynamics can be given as

$$\begin{aligned}\dot{x}_1 &= (A_{11} - A_{12}c_1^T) x_1 \\ x_2 &= -c_1^T x_1.\end{aligned}$$

Remarks

- Sliding function parameter c_1 is such that all the eigenvalue of the matrix $A_{11} - A_{12}c_1^T$ have negative real part.
- The vector c_1 is designed from reduced dynamics.

Sliding Mode Design Using Full Order Lyapunov Matrix

Here, we show how the switching function parameters are designed from the full-order system.

Theorem

^aSuppose that a matrix $Q > 0$ is arbitrarily chosen and $L \in \mathbb{R}^{m \times n}$ is given such that

$$(A - BL)^T P + P(A - BL) + Q = 0 \quad (1)$$

for some $P > 0$. Then, the following statements hold:

- Sliding function parameter c_1 exists of the form

$$c_1^T = P_{22}^{-1} P_{12}^T$$

for a P satisfying (1) where $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^T & P_{22} \end{bmatrix}$.

- For any $P > 0$ satisfying (1), the sliding function parameter $c_1^T = P_{22}^{-1} P_{12}^T$ is a stable sliding function parameter for the system.

^aK.-S. Kim and Y. Park, "Sliding mode design via quadratic performance optimization with pole-clustering constraints", *SIAM Journal of Control and Optimization*, vol. 43, no. 2, pp. 670–684, 2004.

Sliding Mode Design Using Full Order Lyapunov Matrix

Proof of First Statement

Given the sliding function parameter, $c_1^\top$ is a stabilizing gain, and we show that this choice of $P > 0$ guarantees full order system stability. Since $c_1^\top$ is a stabilizing gain, the matrix

$$A_{11} - A_{12}c_1^\top$$

is stable. Define $0 < Q_r = [I_{n-m} - c_1]Q[I_{n-m} - c_1]^\top$. Then there exists $P_r > 0$ such that

$$(A_{11} - A_{12}c_1^\top)^\top P_r + P_r(A_{11} - A_{12}c_1^\top) + Q_r = 0.$$

Fix an arbitrary $P_{22} > 0$. Then let

$$P_{12} = c_1 P_{22}, \quad P_{11} = P_r + P_{12}^\top P_{22}^{-1} P_{12}, \quad L = [L_1 \ L_2]$$

where

$$L_1 = B_2^{-1} \{A_{21} + P_{22}^{-1} P_{12}^\top A_{11} + P_{22}^{-1} A_{12}^\top P_r\}$$

$$L_2 = B_2^{-1} \left\{ A_{22} + P_{22}^{-1} P_{12}^\top A_{12} + \frac{\epsilon}{2} P_{22}^{-1} \right\}, \quad \epsilon > 0.$$

Sliding Mode Design Using Full Order Lyapunov Matrix

Construct $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{12}^\top & P_{22} \end{bmatrix}$ using the pre-designed matrices. Then, with some manipulations, the following relation results

$$T \left((A - BL)^\top P + P(A - BL) + Q \right) T^\top = \begin{bmatrix} R & 0 \\ 0 & -\epsilon I_m \end{bmatrix} = 0$$

where $R = (A_{11} - A_{12}c_1^\top)^\top P_r + P_r(A_{11} - A_{12}c_1^\top) + Q_r$ and $T = \begin{bmatrix} I_{n-m} & -P_{12}P_{22}^{-1} \\ 0 & I_m \end{bmatrix}$.

So, the full order system is stable.

Proof of Second Statement

For a P satisfying (1), define $T_r = [I_{n-m} \quad -P_{12}P_{22}^{-1}]$. Pre and post multiplying T_r and $T_r^\top$ on both sides of (1), gives

$$(A_{11} - A_{12}P_{22}^{-1}P_{12}^\top)^\top P_r + P_r(A_{11} - A_{12}P_{22}^{-1}P_{12}^\top) + T_r Q T_r^\top = 0$$

where $P_r = P_{11} - P_{12}P_{22}^{-1}P_{12}^\top$ which is a positive definite matrix since $P > 0$. Thus by selecting $c_1^\top = P_{22}^{-1}P_{12}^\top$, the stability of the matrix is ensured, and sliding mode enforces stable motion in the system.

Sliding Mode Design Using Full Order Lyapunov Matrix

- From the above result, we see that the sliding function parameter can be chosen from the Lyapunov equation of the full-order system.
- Sliding mode is enforced in the system by the control law

$$u = -(c^\top B)^{-1}(c^\top Ax + K \text{sign}s)$$

where K is a diagonal matrix and $\text{sign}s = [\text{sign}s_1 \ \cdots \ \text{sign}s_m]^\top$.

Sliding Hyperplane Design Under Parametric Uncertainties

Consider an uncertain system^a

$$\dot{x} = (A + \Delta A)x + Bu + Bd$$

where ΔA represents real parametric uncertainties of the form

$$\Delta A = MD(t)N.$$

Here, $M, N^T \in \mathbb{R}^{n \times h}$, and $\mathbb{R}^{h \times h} \ni D(t) = \text{diag}\{\delta_1 I_{r_1}, \dots, \delta_p I_{r_p}\}$ s.t. $|\delta_i| \leq 1$ for $i = 1, 2, \dots, p$ and $t \geq 0$. To see how to write ΔA as $\Delta A = MD(t)N$, we use rank factorization.

- Consider any matrix $S \in \mathbb{R}^{m \times n}$ of rank r . Then, it can be written as $S = S_1 S_2$ where $S_1 \in \mathbb{R}^{m \times r}$ and $S_2 \in \mathbb{R}^{r \times n}$.
- Write ΔA as $\Delta A = \sum_{i=1}^p \delta_i(t) E_i$ with rank of E_i as r_i .

^aK.-S. Kim, Y. Park and S.-H. Oh, "Designing robust sliding hyperplanes for parametric uncertain systems: a Riccati approach", *Automatica*, vol. 36, no. 7, pp. 1041–1048, 2000.

Sliding Hyperplane Design Under Parametric Uncertainties

Then, from rank factorization, we express $E_i = \underline{E}_i \bar{E}_i$ where $\underline{E}_i$ and $\bar{E}_i$ are matrices of appropriate dimensions. So,

$$\begin{aligned}\Delta A &= \underline{E}_1 \delta_1 \bar{E}_1 + \cdots + \underline{E}_p \delta_p \bar{E}_p \\ &= \begin{bmatrix} \underline{E}_1 & \cdots & \underline{E}_p \end{bmatrix} \begin{bmatrix} \delta_1 I_{r_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \delta_p I_{r_p} \end{bmatrix} \begin{bmatrix} \bar{E}_1 \\ \vdots \\ \bar{E}_p \end{bmatrix} = MDN.\end{aligned}$$

Example: Consider ΔA given below with $|\delta_i| \leq 1$ for all $i = 1, 2, 3$ as

$$\begin{aligned}\Delta A &= \begin{bmatrix} 0 & \delta_3 & \delta_1 \\ \delta_1 & 2\delta_1 & \delta_2 \\ 0 & \delta_2 & 0 \end{bmatrix} = \delta_1 \begin{bmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \delta_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \delta_3 \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= \delta_1 E_1 + \delta_2 E_2 + \delta_3 E_3.\end{aligned}$$

Example

To write $E_1 = \underline{E}_1 \bar{E}_1$, we first convert E_1 in reduced row echelon form (i.e., all leading coefficient is one in row echelon form) as

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then, $\underline{E}_1$ is obtained by removing all non-pivotal columns from E_1 and $\bar{E}_1$ is obtained by eliminating all zero rows from the reduced row echelon form of E_1 . This is given as

$$E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{E}_1 \bar{E}_1.$$

Similarly, E_2 can be written in row echelon form as

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Example

It can be verified as

$$\underline{E}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \underline{E}_2 \bar{E}_2$$

$$\underline{E}_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \underline{E}_3 \bar{E}_3.$$

So, we write ΔA as

$$\begin{aligned} \Delta A &= \delta_1 \underline{E}_1 \bar{E}_1 + \delta_2 \underline{E}_2 \bar{E}_2 + \delta_3 \underline{E}_3 \bar{E}_3 = \begin{bmatrix} \underline{E}_1 & \underline{E}_2 & \underline{E}_3 \end{bmatrix} \begin{bmatrix} \delta_1 \bar{E}_1 \\ \delta_2 \bar{E}_2 \\ \delta_3 \bar{E}_3 \end{bmatrix} \\ &= \begin{bmatrix} \underline{E}_1 & \underline{E}_2 & \underline{E}_3 \end{bmatrix} \begin{bmatrix} \delta_1 h_2 & 0 & 0 \\ 0 & \delta_2 h_2 & 0 \\ 0 & 0 & \delta_3 h_1 \end{bmatrix} \begin{bmatrix} \bar{E}_1 \\ \bar{E}_2 \\ \bar{E}_3 \end{bmatrix} = MDN. \end{aligned}$$

Sliding Hyperplane Design Under Parametric Uncertainties

Definition

A system is said to be quadratically stable if there exists a linear state feedback control law $u = -Lx$, $P > 0$ and a scalar $\alpha > 0$ such that $V = x^T P x$ is Lyapunov function and $\dot{V} < -\alpha \|x\|^2$.

Theorem

Let $d = 0$. The system is quadratically stable with linear state feedback control $u = -Lx$ if there exists $P > 0$, $X \in S_{\text{sym}}$ and $U \in S_{\text{skew}}$ such that

$$(A - BL)^T P + P(A - BL) + Q + PMXM^T P + (N + PMU^T)X^{-1}(N + PMU^T)^T < 0$$

for some $Q \geq 0$ and

$$S_{\text{sym}} = \{X : XD(t) = D(t)X, X > 0\}$$

$$S_{\text{skew}} = \{U : UD(t) = D(t)U, U = -U^T\}.$$

Sliding Hyperplane Design Under Parametric Uncertainties

Proof

Consider $V = x^\top P x$. Taking time derivative, we obtain

$$\dot{V} = x^\top \{(A - BL)^\top P + P(A - BL)\}x + x^\top \{PMDN + N^\top D^\top M^\top P\}x.$$

The second term can be written as

$$\begin{aligned} PMDN + N^\top D^\top M^\top P &= PMDH + H^\top D^\top M^\top P \\ &= (PMX^{1/2})(DX^{-1/2}H) + (DX^{-1/2}H)^\top (PMX^{1/2})^\top \\ &\leq PMXM^\top P + H^\top X^{-1}H \end{aligned}$$

where $H = N + UM^\top P$. The first equality holds since $U \in S_{\text{skew}}$ and the second equality holds because $X \in S_{\text{sym}}$, so $D = X^{1/2}DX^{-1/2}$. In the last step, we use the bounding inequality given as

$$HY + Y^\top H^\top \leq HH^\top + Y^\top Y.$$

So, combining all these, we write

$$\dot{V} \leq x^\top \{(A - BL)^\top P + P(A - BL) + PMXM^\top P + H^\top X^{-1}H\}x.$$

Sliding Hyperplane Design Under Parametric Uncertainties

Proof

In other words, if

$$(A - BL)^T P + P(A - BL) + Q + PMXM^T P + H^T X^{-1} H < 0$$

then the system is quadratically stable for some $Q \geq 0$.

Theorem

Sliding mode exists in the system if there exists $P > 0$, $L \in \mathbb{R}^{m \times n}$, $X \in S_{\text{sym}}$ and $U \in S_{\text{skew}}$ such that

$$(A - BL)^T P + P(A - BL) + Q + PMXM^T P + (N + PMU^T)X^{-1}(N + PMU^T)^T < 0 \quad (2)$$

for some $Q \geq 0$.

Proof

Pre and post multiplying both sides of (2) by T_r and T_r^T where $T_r = [I_{n-m} \quad -P_{12}P_{22}^{-1}]$. Then, after some simple calculation it gives $c_1^T = P_{22}^{-1}P_{12}^T$ such that reduced dynamics is stable.

Thank You!